

DengueCAD

System Architecture

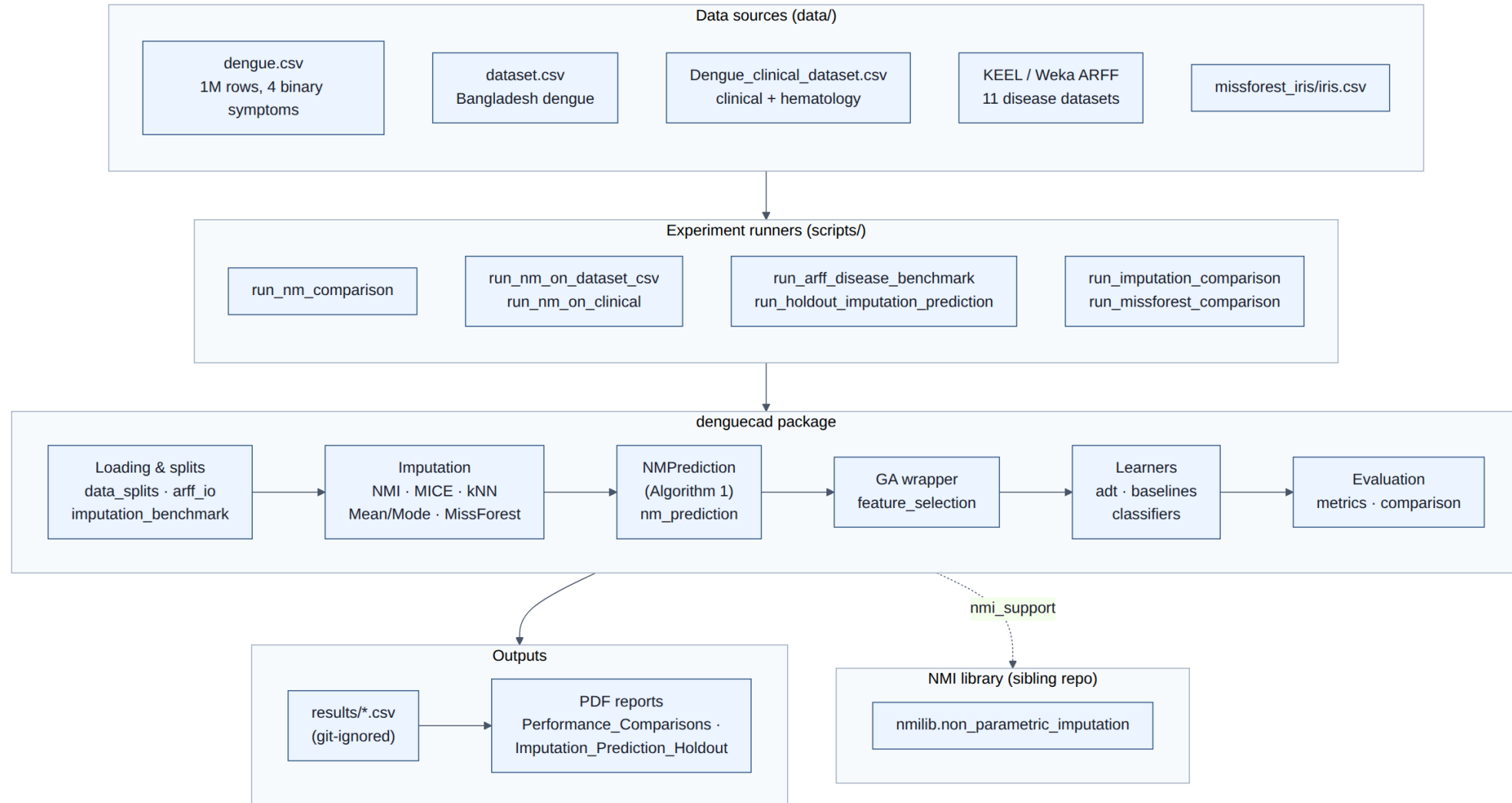
NMI imputation + NMPrediction (IEEE TITB 2012, Algorithm 1) for computer-aided diagnosis

October 2026

What DengueCAD does

- Implements NMPrediction (Rao & Kumar, IEEE TITB 2012)
 - NMI imputation → GA wrapper feature selection → stratified 10-fold ADT
 - Reports Accuracy, AUC, Sensitivity (SE), Specificity (SP)
- Compares NM against C4.5, Logistic Regression (LOR) and SVM (RBF)
 - Table I performance · Table II Wilcoxon test · Table III influential features
- Benchmarks imputation: NMI vs MissForest, MICE, kNN, Mean/Median/Mode
 - Hold-out experiment: imputation accuracy and hidden-decision prediction accuracy
- Disease-agnostic: dengue CSVs plus 11 KEEL / Weka ARFF disease datasets
- Reproducible: scripted experiments, generated PDF report, CI on every push

System overview

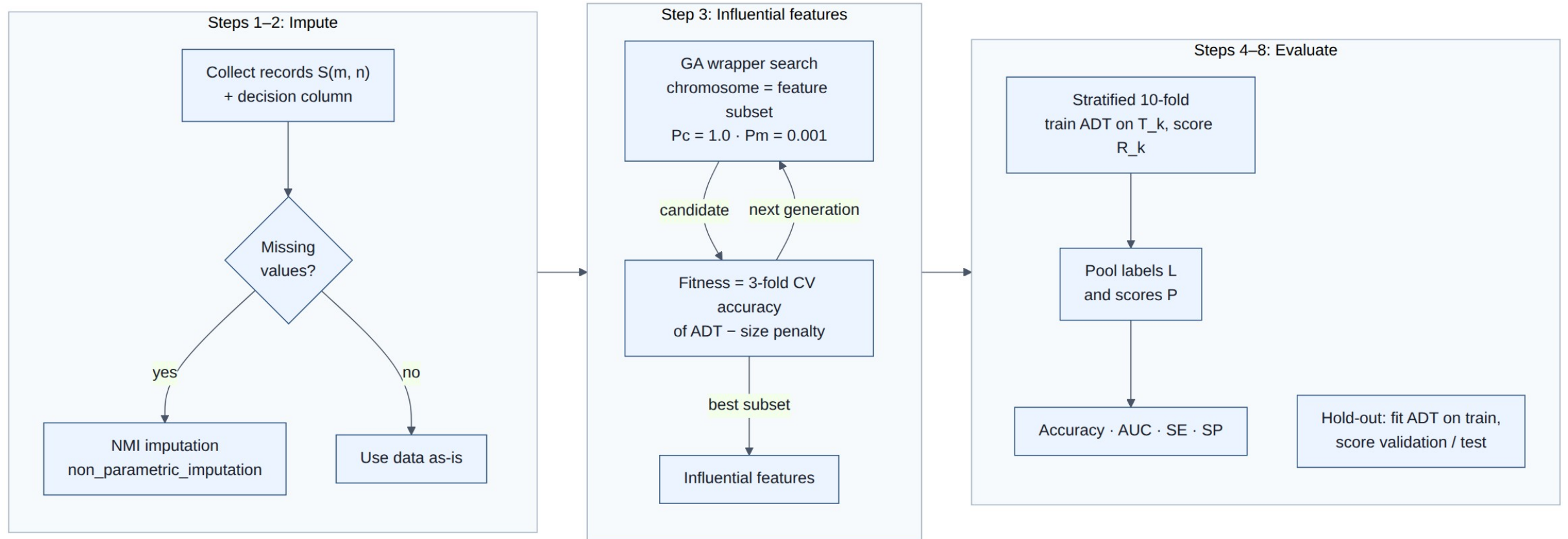


Data sources → experiment scripts → denguecad package → CSV results and PDF report; imputation delegates to the sibling NMI library.

Package components denguecad/

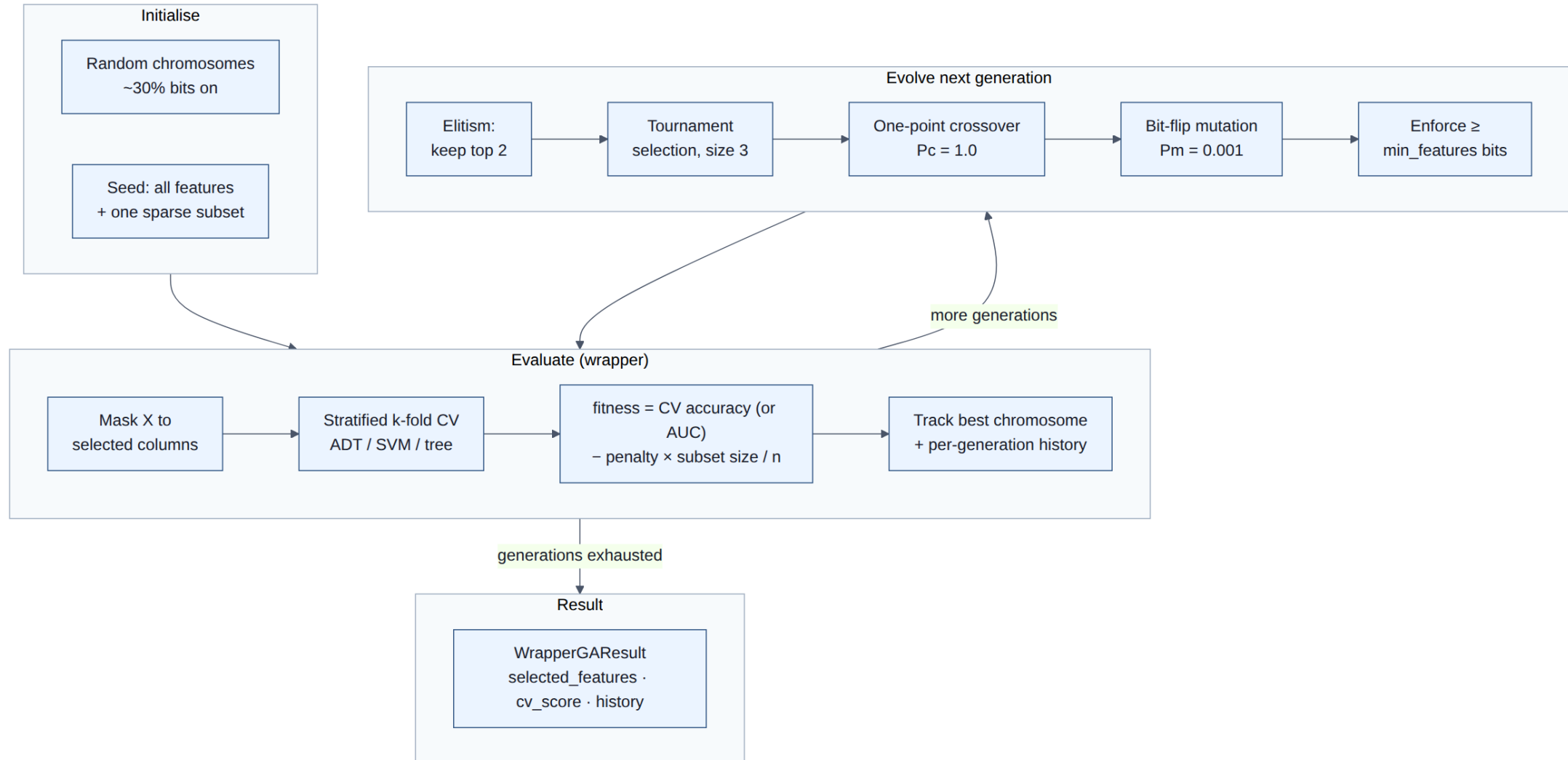
Layer	Modules	Responsibility
Data	data_splits · arff_io · imputation_benchmark	Load, encode, split (70k / 10k / 20k), ARFF with KEEL ranges
Imputation	nmi_support · imputation_benchmark · missforest_impute	NMI, MICE, kNN, Mean/Mode, MissForest
Algorithm	nm_prediction · feature_selection	NMPrediction (Algorithm 1), GA wrapper search
Learners	adt · baselines · classifiers	ADT, C4.5, LOR, SVM (RBF), classifier factory
Evaluation	metrics · comparison · holdout_experiment	Accuracy/AUC/SE/SP, Tables I–III, Wilcoxon, hold- out protocol
Reporting	build_comparison_pdf · build_holdout_comparison_pdf	Performance and hold-out PDF reports with highlighted best results

NMPrediction pipeline Algorithm 1



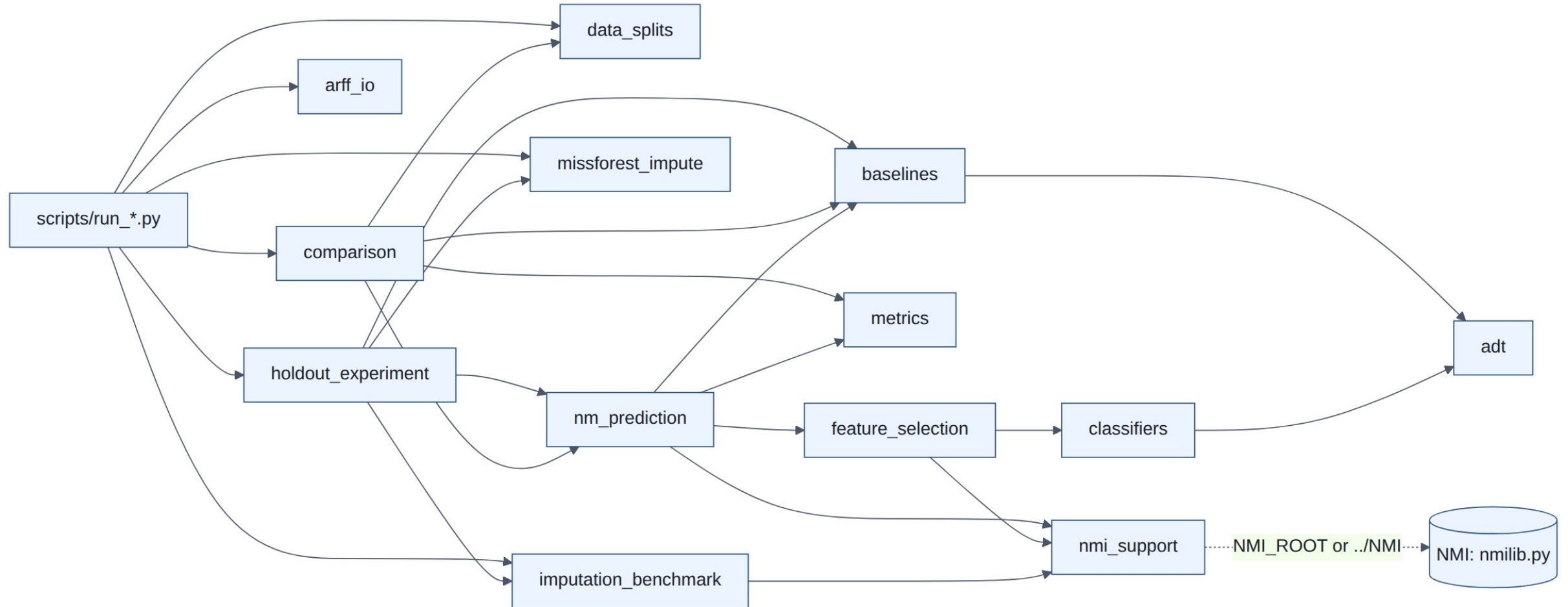
Algorithm 1: impute missing values with NMI, pick influential features with a GA wrapper, then score ADT with stratified 10-fold CV.

GA wrapper feature selection Section C



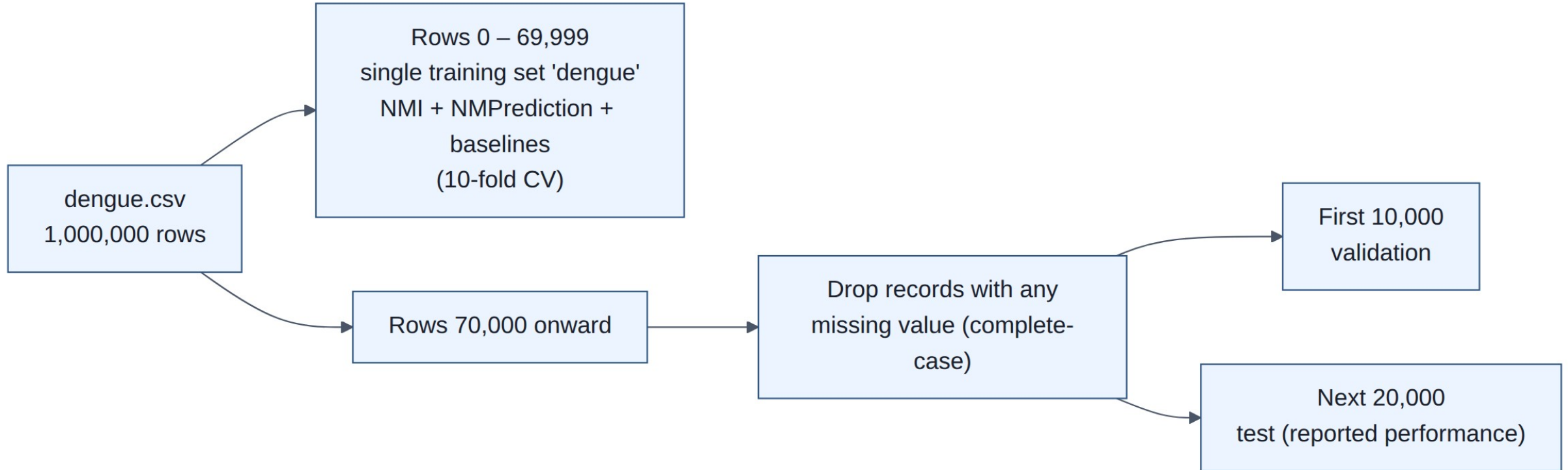
Kohavi & John wrapper: the classifier scores each subset. $P_c = 1.0$, $P_m = 0.001$, elitism 2, tournament 3, small subset-size penalty.

Module dependencies



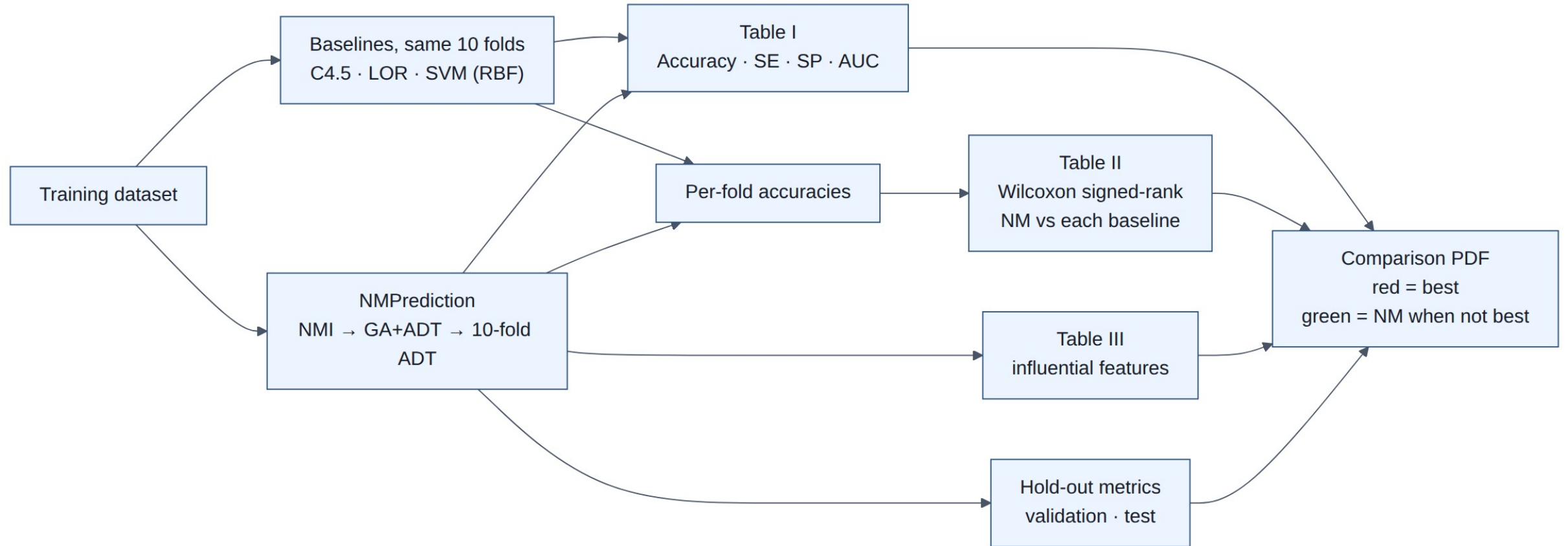
Arrows point to imported modules. Only nmi_support touches the external NMI library.

Data protocol: dengue.csv



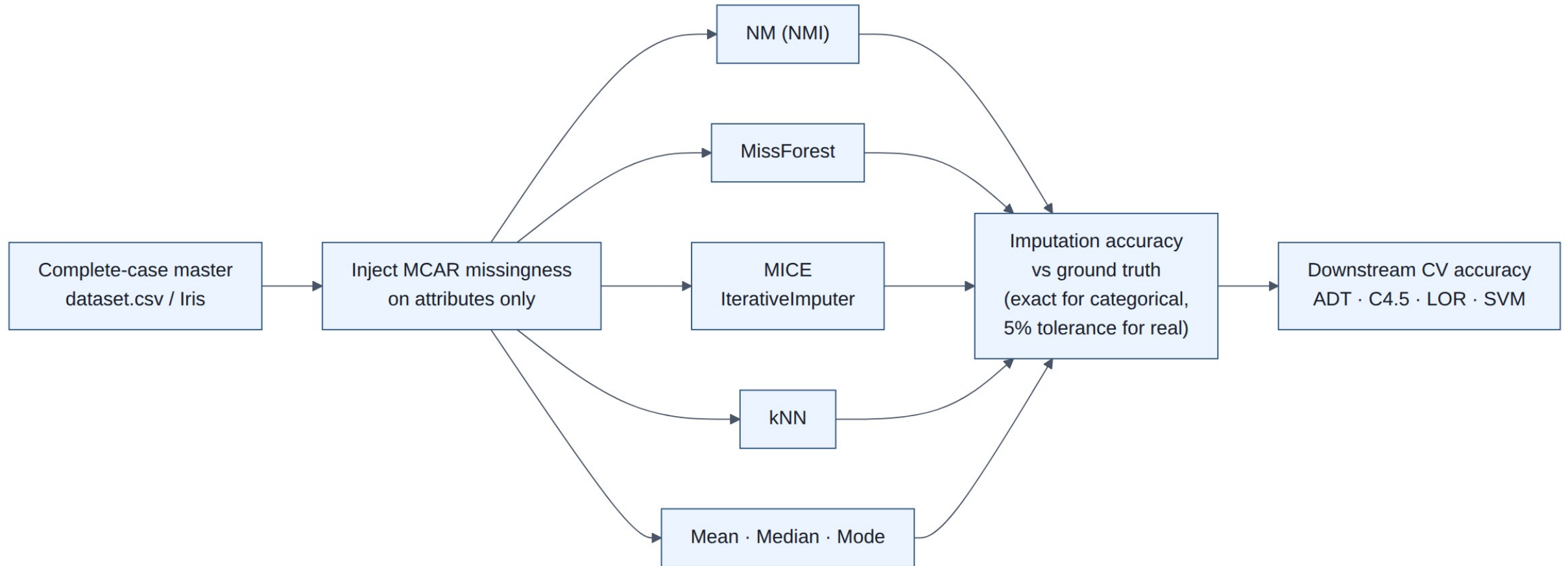
70,000-row single training set · 10,000 complete-case validation · 20,000 complete-case test.

NM vs baselines evaluation



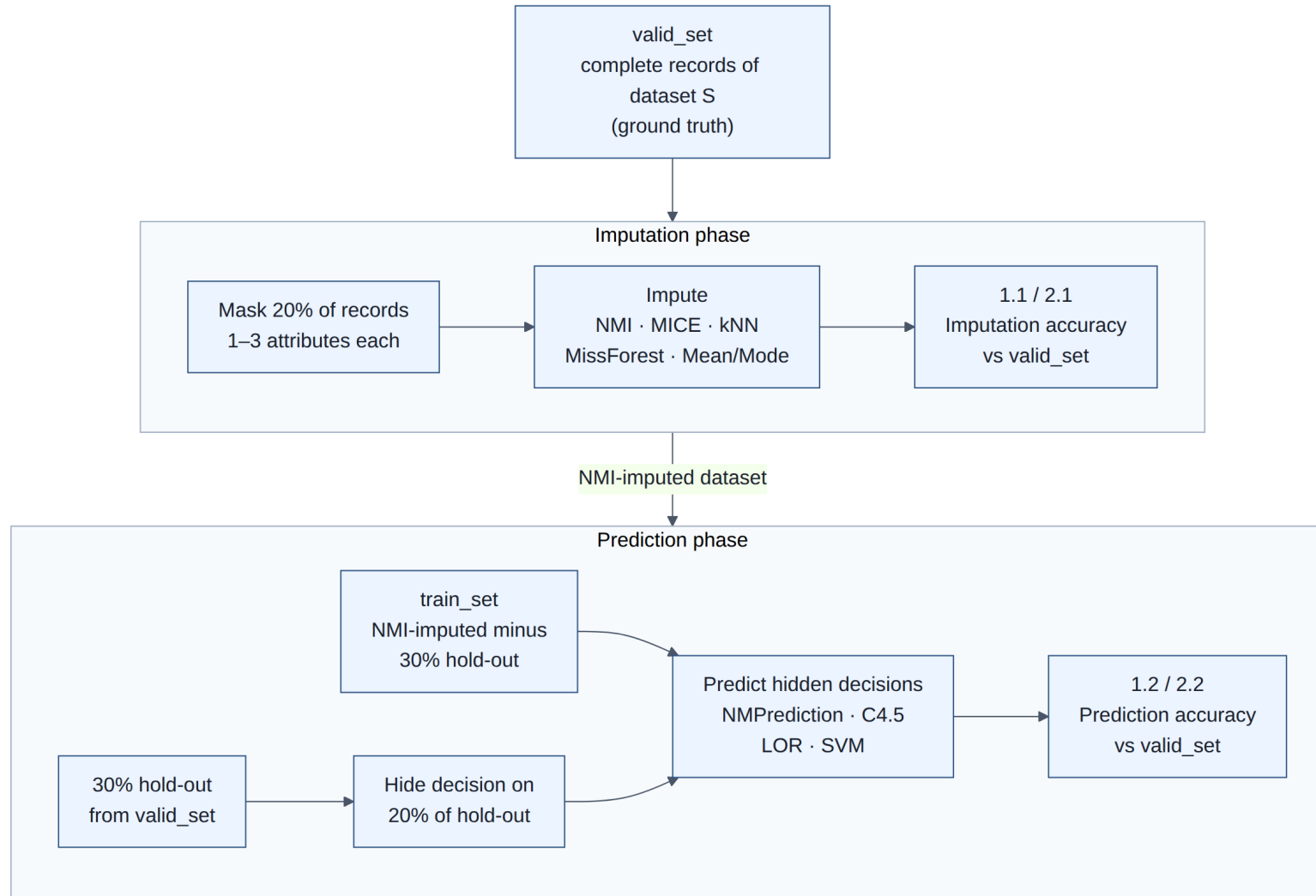
All methods share the same folds; per-fold accuracies feed the Wilcoxon signed-rank test.

Imputation benchmark



Mask a complete dataset with MCAR missingness, impute with each method, score against ground truth and downstream classifiers.

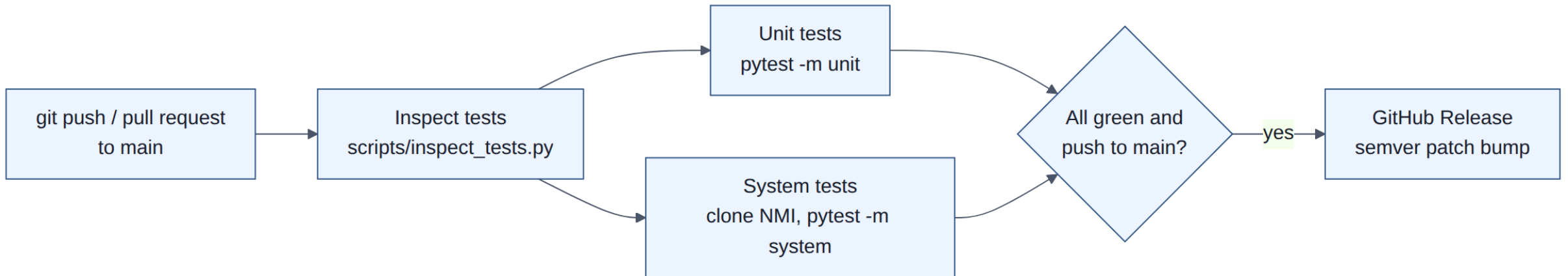
Imputation + prediction hold-out experiment holdout_experiment.py



valid_set kept aside · 20% of records masked and imputed · NMI-imputed train_set · decisions hidden on 20% of a 30% hold-out and predicted.

Experiment runners and outputs scripts/ → results/

Script	Dataset	Output
run_nm_comparison.py	dengue.csv (70k / 10k / 20k)	Tables I–III, hold-out metrics, split summary
run_nm_on_dataset_csv.py	dataset.csv (Bangladesh)	results/nmprediction_dataset_csv/
run_nm_on_clinical.py	Dengue_clinical_dataset.csv	results/nmprediction_clinical/
run_arff_disease_benchmark.py	11 disease ARFFs	results/arff_disease_comparison/
run_imputation_comparison.py	dataset.csv + MCAR	results/imputation_dataset_csv/
run_missforest_comparison.py	Iris + 20% MCAR	results/missforest_comparison/
run_holdout_imputation_prediction.py	5 dengue sets + 11 ARFFs	results/holdout_imputation_prediction/
build_comparison_pdf.py	all results	Performance comparison PDF (docs/)
build_holdout_comparison_pdf.py	hold-out results	Imputation / prediction hold-out PDF (docs/)



GitHub Actions on every push and pull request; a green push to main cuts a semver release.

Design decisions and extension points

- NMI kept as an external library behind nmi_support
 - Swap the checkout with NMI_ROOT; no code changes
- Pluggable learners through factories (baselines, classifiers)
 - GA wrapper can evaluate with ADT, SVM or decision tree
- Disease-agnostic data layer
 - New datasets: add a CSV/ARFF loader, reuse run_dataset_comparison
- Reproducibility
 - Fixed random seeds, scripted runs, PDF and diagrams regenerated from source

Running and regenerating

```
source .venv/bin/activate
python scripts/run_nm_comparison.py           # dengue.csv experiment
python scripts/run_arff_disease_benchmark.py  # multi-disease ARFFs
python scripts/run_holdout_imputation_prediction.py # hold-out experiment
python scripts/build_comparison_pdf.py        # performance report
python scripts/build_holdout_comparison_pdf.py # hold-out report
python scripts/render_architecture_diagrams.py # diagrams
python scripts/build_architecture_slides.py --pdf # this deck
bash scripts/ci_local.sh                     # CI gates before push
```